Jiyon PJ

ENERGY ENGINEER | AI FOR ENERGY SYSTEMS

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PROFILE SUMMARY

Mechanical Engineer with a recently completed Master's in Energy Engineering and Management, complemented by 2 years of industry experience in engineering project coordination prior to Master's studies. Completed 24+ permit applications with 80% first-pass approval and reduced approval timelines by 40%. Experienced in engineering analysis and drawing interpretation using Python, MATLAB, Excel and AutoCAD. Developed hands-on experience in building AI-driven engineering tools, including Retrieval-Augmented Generation (RAG) systems, LLM integration and data pipeline development through applied projects. Seeking an early-career engineering role to apply and further develop domain expertise.

KEY SKILLS

Mechanical Engineering Fundamentals

- Thermodynamics, Heat Transfer, Fluid Mechanics, HVAC Systems, Heat Exchangers, Refrigeration & Heat Pump Systems, Performance Analysis of Energy Systems

Energy & Sustainability

- Renewable Energy Systems, Hydrogen Systems, Energy Efficiency Analysis, Industrial Energy Conservation, Decarbonization Principles, ESG Fundamentals

Technical Tools

- AutoCAD, SolidWorks, Autodesk Fusion, Python, MATLAB, Simulink, Microsoft Excel, COMSOL Multiphysics

AI Systems Engineering

- Semantic Search, RAG, LLM Integration, Agentic Systems, Pinecone (Vector Database), SvelteKit, Gemini API, ETL/ELT Pipelines, Data Preprocessing

Engineering Project Coordination

- Technical Documentation, Engineering Drawing Interpretation, Regulatory Compliance, Site Coordination, Workflow & Submission Management, Stakeholder Coordination

EDUCATION

National Institute of Technology Calicut (NITC)

M.Tech – *Energy Engineering and Management* **CGPA**: 8.29 *July* 2023 – *August* 2025

Master's Thesis: Experimental and Computational Studies on Hydrogen Embrittlement

- Major responsibilities included specimen acquisition, preparation and conducting laboratory tests under strict safety protocols and reporting the results in structured technical formats to industrial and academic reviewers.
- Performed tensile testing and scanning electron microscopy (SEM) to analyse fracture mechanisms and assisted in COMSOL Multiphysics-based simulations.
- Developed Python and MATLAB scripts for quantitative analysis of stress – strain datasets to derive meaningful results.
- Presented research findings at the Global Conference on Decarbonizing India (2025) alongside the PhD scholar supervising my work.

College of Engineering Trivandrum (CET)

B.Tech – *Mechanical Engineering* **CGPA**: 8.03 2017–2021

Final Year Project: Forced Convection Heat Transfer Simulation

- Major responsibilities included leading the project team during COVID-19 lockdown constraints to define and deliver the project within the practical limitations.
- Developed a MATLAB Simulink model to analyse forced convection heat transfer under varying thermal conditions.
- Validated simulation outputs against theoretical correlations, maintaining deviations within ±10% across relevant operating conditions.

PROJECTS

AITfES – A modular RAG-based AI system

Designed and deployed a prototype of a modular AI system called "***AI Troubleshooting for Energy Systems***" (AITfES) to assist engineering decision making using Retrieval-Augmented Generation (RAG) for specified domains. Built automated data pipelines scaling structured data from 148 to 12,000+ records for semantic retrieval. Integrated LLM, vector database and frontend components using Gemini API, Pinecone and SvelteKit into a customizable full-stack system.

- Engineered a **domain-constrained diagnostic RAG pipeline** to support structured troubleshooting of energy systems.
- Integrated **multi-model fallback orchestration** to maintain reliability during troubleshooting sessions, ensuring uninterrupted diagnostic support under API failures or rate limits.
- Implemented **strict scope enforcement and rejection rules** to prevent out of domain queries, ensuring responses are actionable diagnostic steps rather than conversational outputs.

Live URL: aitfes.jiyon.online | **GitHub:** github.com/thenewlegend/AITfES

WORK EXPERIENCE

DAZZLE: Designers, Engineers & Contractors | Kerala, India

Project Coordinator (*Engineering Projects*) 2021 – 2023

- Achieved an 80% first-pass approval rate across 24 engineering permit applications by preparing accurate, regulation-compliant applications aligned with KPBR (Kerala Panchayat Building Rules).
- Interpreted engineering drawings using AutoCAD and verified compliance with regulatory frameworks to support accurate project execution.
- Improved coordination and submission processes, contributing to an estimated 40% reduction in approval timelines. Coordinated with engineers, clients and regulatory authorities.
- Assisted in site supervision, identifying and addressing on-site deviations to ensure compliance with approved plans and KPBR standards.

PROFESSIONAL CERTIFICATE

Career Essentials in Sustainable Tech (Microsoft & LinkedIn)

- Gained practical exposure to SDGs, circular economy models, and resilience planning, thereby enhancing the ability to apply sustainability principles to industry challenges.
- Established a solid foundation in sustainability concepts within a technology-driven context.